

How AI data centers are affecting wildlife

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Source Content

The computers behind artificial intelligence (AI) can have real consequences for animals. As thousands of new data centers are built around the world, scientists are studying how they affect wildlife and the environment. Here are some of the main ways AI data centers can impact animals.

Habitat loss One of the biggest concerns is where new data centers are built. Large AI data centres can cover many hectares of land. Before construction begins, forests, grasslands, wetlands or farmland may be cleared to make room for buildings, roads and electricity infrastructure. This destroys wildlife habitat and split ecosystems into smaller areas, making it harder for animals to find food, mates and safe places to live. Scientists call this habitat fragmentation, and it is considered one of the biggest threats to biodiversity worldwide.

Study: Environmental impact and net-zero pathways for sustainable AI servers Many AI data centers require new electricity infrastructure. Building transmission lines and substations can fragment habitats and increase the risk of bird collisions with power lines. New roads built for construction and maintenance may also make previously undisturbed animal habitats easier for humans to access.

Water use AI servers generate large amounts of heat and must be cooled. Many data centers use millions of liters of freshwater every year for cooling systems. In regions already facing drought, this can reduce the amount of water available for rivers, wetlands and wildlife. Lower water levels can affect fish, amphibians, birds and mammals that rely on healthy freshwater ecosystems. Researchers also warn that warm wastewater released from cooling systems can affect aquatic environments if it is not managed properly.

Study: Data Centers and Water: Challenges and Solutions for Sustainable Cooling Greenhouse gas emissions Data centers require enormous amounts of electricity. According to the International Energy Agency (IEA), electricity demand from data centers is increasing rapidly as AI expands. If that electricity comes from fossil fuels, it contributes to greenhouse gas emissions. If data centres use electricity from coal, oil or gas, they produce greenhouse gases. These gases trap heat in the atmosphere, making the Earth warmer. A warmer planet can affect animals by changing their habitats, causing more heatwaves, droughts and wildfires.

Study: The carbon and water footprints of data centers and what this could mean for artificial intelligence Noise and light Although data centers appear quiet from the outside, they often operate 24 hours a day. Cooling equipment, backup generators and bright security lighting can disturb wildlife, especially nocturnal animals such as bats, owls and insects. Artificial light can also confuse migrating birds and attract insects away from natural habitats. While more research is still needed, ecologists say these local impacts should be considered when choosing sites for new facilities. Researchers are calling for greater transparency from technology companies about the environ-

mental costs of AI, including its water use, carbon emissions and effects on biodiversity. Can data centres become more wildlife-friendly? Not all data centres have the same impact on the environment. Experts say companies can reduce harm by building on land that has already been developed instead of clearing natural habitats. They can also use renewable energy, choose cooling systems that use less freshwater, recycle water and restore wildlife habitats around their facilities. Carefully selecting locations away from sensitive ecosystems can also help protect animals. Balancing AI and nature Artificial intelligence is expected to continue growing for many years. At the same time, conservationists say protecting wildlife will require careful planning as more data centres are built. Many scientists believe the challenge is not whether AI should exist, but how its infrastructure can be developed in ways that minimise impacts on animals, water resources and natural habitats while still supporting technological progress.